Object Oriented Programming

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Last Week

- Project
- Recap
- Overloading vs. Overriding
- Levels of Inheritance
- Multiple Inheritance

This Week

- Quiz
- Assignment
- Polymorphism
  - Virtual Functions
  - Pure Virtual Functions
  - Abstract Base Classes

Motivation

- Shape → Rectangle, Triangle
- Vehicle → Car, Truck
- Weapon → Bomp, Gun

Virtual Function

- A virtual member is a member function that can be redefined in a derived class
- A class that declares or inherits a virtual function is called a polymorphic class
- Objects can also be instantiated from the Base (Polymorphic class)
- Polymorphism in C++ is accomplished by allowing pointers

Virtual Function

```
class Polygon {
  protected:
    int width, height;
  public:
    void set_values (int a, int b)
      { width=a; height=b; }
    virtual int area ()
      { return 0; }
};
```

```
class Rectangle: public Polygon {
  public:
    int area ()
      { return width * height; }
};

class Triangle: public Polygon {
  public:
    int area ()
      { return (width * height / 2); }
};

int main () {
  Rectangle rect;
  Triangle trgl;
  Polygon poly;
  Polygon * ppoly1 = &rect;
  Polygon * ppoly2 = &trgl;
  Polygon * ppoly3 = &poly;
  ppoly1->set_values (4,5);
  ppoly2->set_values (4,5);
  ppoly3->set_values (4,5);
  cout << ppoly1->area() << '\n';
  cout << ppoly2->area() << '\n';
  cout << ppoly3->area() << '\n';
  return 0;
}
```

Polymorphism

- Many shapes
- polymorphism is the provision of a single interface to entities of different types
- Or the use of a single symbol to represent multiple different types
- Types:
  - Compile Time (Templates and Generic Programming)
  - Run Time (Virtual functions)

Abstract Base Classs

- Have virtual functions without definitions
- Virtual functions without definitions are called pure virtual functions
- Cannot be instantiated

Abstract Base Class

```
// abstract class CPolygon
class Polygon {
  protected:
    int width, height;
  public:
    void set_values (int a, int b)
      { width=a; height=b; }
    virtual int area () =0;
};
```

```
class Rectangle: public Polygon {
  public:
    Rectangle(int a,int b) : Polygon(a,b) {}
    int area()
      { return width*height; }
};

class Triangle: public Polygon {
  public:
    Triangle(int a,int b) : Polygon(a,b) {}
    int area()
      { return width*height/2; }
};

int main () {
  Polygon * ppoly1 = new Rectangle (4,5);
  Polygon * ppoly2 = new Triangle (4,5);
  ppoly1->printarea();
  ppoly2->printarea();
  delete ppoly1;
  delete ppoly2;
  return 0;
}
```

Summary

- Virtual members and abstract classes grant C++ polymorphic characteristics, most useful for object-oriented projects